Temas Documentation

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rubu

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CHAPTER

ONE

PRODUCT DESCRIPTION

The **Temas** product is a mini gimbal with multiple functionalities, powered by built-in components. These components enable various operations such as controlling a camera, lidar laser, and two motors via a Wi-Fi interface. A supplied software suite allows for the following features:

- Motor movement
- Live camera stream
- Distance measurement
- 3D view generation
- Point cloud creation
- Object tracking
- Object position control
- QR code scanning

An API is also available to allow you to implement custom solutions in the fields of artificial intelligence, machine vision, and deep learning. The object tracking feature enables 3D trajectory recording and evaluation, which aligns with modern digital-age physical experiments.

The Temas product integrates seamlessly into Industry 4.0 process optimization, offering all these features and more!

Important: For more advanced use cases and API documentation, please refer to the API documentation.

CHAPTER

TWO

FEATURES OVERVIEW

- **Industry 4.0 integration**: Temas is designed for process optimization.
- **Customizable**: Temas offers an API for building custom AI, machine vision, and deep learning applications.
- **Versatile functionalities**: With built-in components, it enables real-time monitoring and control.

2.1 Quick reference for Temas

This section covers the use of the Temas modules.

2.1.1 Import

First you need to import the module into your project. To do that you need to download the package from pypi. Download Package:

Note: Download https://beispiellink.com/package

Install via PyPi:

Note: pip install https://beispiellink.com/package

After download and installing the package import it into your project

2.1.2 Tutorial

Import Package

```
from rubu import temas
```

Initialize Control

```
control=temas.Control()
```

Initialize Common

```
common=temas.Common()
```

2.1.3 Examples

Example Code for your project. More information about modules is in *Temas module*.

Get distance

```
dis=control.distance()
dis_mean=control.mean_distance()
```

Move to start position or any other position

```
control.move_home()
control.move_left_coarse()
control.azimutal_position(1.1)  #1.1°
control.elevation_position(1.1) #1.1°
```

Get Temperature

```
temperature=common.get_temperature()
```

Get Ip address

```
ip_address=common.get_ip()
```

Initialize Frame Start camera

```
...
control=temas.Control()
control.move_home()
camera=temas.Camera()
camera.start_thread()
...

while True:
   try:
      frame = camera.get_frame()
      ...
      ...
         cv2.imshow("Image",frame)
   except:
      pass

   if cv2.waitKey(1) == ord('q'):
      break

out.release()
camera.stop_thread()
```

Get Port of camera

```
port=common.get_port_camera()
```

Live streaming in web browser

```
http://ip:8081
```

Shutdown or restart system

```
common.shutdown() //shutdown
common.restart()  //restart
```

2.2 Technical specification

Table 1: specification

parameter	value
Elevation	-80° to 80°
Azimuthal	-70° to 90°
Power supply	5 V/3 A DC (USB Type C)
Operating time at 4.8 V	0.09 s/60°
Gearbox	Titan
Weight	—pending—
Operating temperature	Room temperature
Camera Sensor	Sony IMX219
Image resolution	3208 x 2465 Pixel (8 MP)
Video resolution	1080p30, 720p60
LiveStream resolution (supplied Software)	320 x 240 Pixel
Pixel pitch	1.12 µm
Focal lenght	3.04 mm
Field of view horizontal	62.2°
Field of view vertikal	48.8°

continues on next page

Table 1 – continued from previous page

parameter	value
Laser class	1
Range	1 m to 40 m
Resoulution	+/- 1 cm
Accuracy	+/- 2.5 cm
Wavelenght	905 nm
Peak performance (power)	1.3 W
Refresh rate	500 Hz
Beam divergence	8 mrad

2.3 Temas module

class temas.**Connect**(*hostname='rubu'*)

Bases: object

This class alows the Temas connection

class temas.**Socket**(*port*)

Bases: object

This class send commands to Temas

send_cmd(*befehl*)

Send commands

Parameters
befehl (*List, optional*) – A command is given

Returns
Data of socket temas.send_cmd

Raises
Exception

class temas.**Camera**(*port=8081*)

Bases: object

This class allows the control of camera

update()

Update Frame

start_thread()

Start thread

stop_thread()

Stop thread

get_frame()

Get frame

Returns
Frame

Return type
Array

class temas.**Control**(*port=8082*)

Bases: object

This class alows the control of motors and laser

distance()

Laser distance measurment

Returns
Value of the distance [cm]

Return type
String

mean_distance()

Mean laser distance measurement with a number of 30

Returns
Value of the mean_distance [cm]

Return type
String

azimutal_position(*degree*)

Set a movement to the azimutal position

Parameters
degree (*Value*) – Value of azimutal movement [°]

Returns
Motor is moved?

Return type
Boolean

elevation_position(*degree*)

Set a movement to the elevation position

Parameters
degree (*Value*) – Value of elevation movement [°]

Returns
Motor is moved?

Return type
Boolean

move_home()

Move to home position

Returns
Motor at start position?

Return type
Boolean

`move_left_coarse()`

Move left in 10° steps

Returns
Value

Return type
String

`move_right_coarse()`

Move right in 10° steps

Returns
Value

Return type
String

`move_up_coarse()`

Move up in 10° steps

Returns
Value

Return type
String

`move_down_coarse()`

Move down in 10° steps

Returns
Value

Return type
String

`move_left_fast()`

Move left in 5° steps

Returns
Value

Return type
String

`move_right_fast()`

Move right in 5° steps

Returns
Value

Return type
String

`move_up_fast()`

Move up in 5° steps

Returns
Value

Return type
String

`move_down_fast()`

Move down in 5° steps

Returns
Value

Return type
String

`move_left()`
Move left in 1° steps

Returns
Value

Return type
String

`move_right()`
Move right in 1° steps

Returns
Value

Return type
String

`move_up()`
Move up in 1° steps

Returns
Value

Return type
String

`move_down()`
Move down in 1° steps

Returns
Value

Return type
String

`move_left_fine()`
Move left in 0.1° steps

Returns
Value

Return type
String

`move_right_fine()`
Move right in 0.1° steps

Returns
Value

Return type
String

`move_up_fine()`
Move up in 0.1° steps

Returns
Value

Return type
String

move_down_fine()

Move down in 0.1° steps

Returns
Value

Return type
String

class temas.**Common**(*port=8083*)

Bases: object

This class included the common settings of Temas

set_resolution1()

Set resolution of camera to 320x240

Returns
Set resolution1

Return type
Boolean

set_resolution2()

Set resolution of camera to 640x480

Returns
Set resolution2

Return type
Boolean

set_resolution3()

Set resolution of camera to 1024x768

Returns
Set resolution3

Return type
Boolean

shutdown()

Shutdown Temas

Returns
Shutdown temas

Return type
Boolean

restart()

Restart Temas

Returns
Restart temas

Return type
Boolean

get_temperature()

Read Temperature of Temas

Returns
Temperature Temas

Return type
String

`get_ip()`

Get ip address of Temas

Returns
Ip address

Return type
String

`get_hostname()`

Get hostname of Temas

Returns
Hostname is returned

Return type
String

`get_port_camera()`

Get port of Temas camera

Returns
Port from camera

Return type
String

`get_port_common()`

Get port of Temas common

Returns
Port from Temas common

Return type
String

`get_port_control()`

Get port of control

Returns
Port from control

Return type
String

`get_ip_hosts()`

Get the host ip address of Temas

Returns
Ip adress of host

Return type
String

`get_mac()`

Get the mac address of Temas

Returns
Returns the mac address

Return type
String

2.4 RUBU license information

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FOUR

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